

SOPHIA D'AGOSTINO RUBENS

INSTRUMENT-AWARE COSMOLOGY GRADUATE STUDENT

 <https://sophiarubens.github.io/web/> |  sophia.rubens@mail.mcgill.ca

 [Sophia Rubens](#) |  [sophiarubens](#)

I am a physics PhD student at McGill University. I am driven to help advance our understanding of the large-scale structure, composition, and evolution of the universe. I devise simulations and analysis techniques to advance next-generation intensity mapping efforts. I also enjoy volunteering in programs aimed at helping members of the general public of all ages foster curiosity about physics, space, and observational astronomy.

EDUCATION

2024—	PhD Physics , McGill University	GPA: 4.00/4.00
	<i>Supervisors:</i> Adrian Liu, Hsin Cynthia Chiang	
2020–2024	AB Physics , Dartmouth College	GPA: 3.95/4.00
	<i>Supervisors:</i> Dong-Jin Kim, John Thorstensen	
	<i>Thesis:</i> Understanding the Gravitationally Lensed Blazar PKS 1830-211	
	<i>Distinctions:</i> High honours in physics, minor in astronomy	

AWARDS & HONOURS

2025	McGill Physics Graduate Excellence (GEF) Scholarship, merit-based	\$5000 CAD
2024	Phi Beta Kappa academic honour society, Dartmouth College chapter	
2024	Sigma Xi science honour society, Dartmouth College chapter	
2024	Dartmouth Physics and Astronomy Chair's Prize	\$500 USD
2024	Christopher Reed Science Competition at Dartmouth College, third place	
2024	First Honor Group at Dartmouth College (Rufus Choate Scholar; top 5%)	
2023	Second Honor Group at Dartmouth College (top 15%)	
2022	Second Honor Group at Dartmouth College (top 15%)	
2022	Presidential Scholar at Dartmouth College	\$1500 USD
2021	Barbara E. Crute Memorial Scholarship at Dartmouth College	\$1200 USD
2021	Third Honor Group at Dartmouth College (top 35%)	
2020	Advanced Placement Scholar Award with Distinction	
2020	Top 10 student at Exeter High School (New Hampshire)	
2020	Award with High Honours, Society of Women Engineers, Exeter High School (NH)	
2020	Latin Achievement Award, Exeter High School (NH)	
2020	New Hampshire Scholar with STEM and arts emphasis	
2019	Harvard Book Award, Harvard Club of Boston	
2019	Social studies writing award, Exeter High School (NH)	
2019	Math achievement award, New Hampshire Teachers of Mathematics	
2016-20	Gold medal on the National Latin Exam (I, II, III, III/IV Prose, III/IV Poetry)	

PUBLICATIONS

- CHORD Collaboration: Bandura, K., [35 more authors], **Rubens, S. D.**, [13 more authors]. "Overview of the Canadian Hydrogen Observatory and Radio Transient Detector (CHORD) Project," 2026, submitted to *Proceedings of SPIE*, <https://arxiv.org/abs/2607.09374>.

SKILLS

Programming: Python (numpy, matplotlib, astropy, pandas...), C, Fortran, MatLab

Computing: bash, git, slurm, MPI, Mathematica, L^AT_EX, AIPS, difmap

Languages: English (native), Spanish (native), Catalan (intermediate), French (intermediate)

TEACHING ASSISTANTSHIPS

PHYS 257	Experimental Methods I	McGill	Fall '26	Michael Hilke
PHYS 102	Introductory Physics—Electromagnetism	McGill	Winter '25	Nikolaos Provatas
PHYS 230	Dynamics of Simple Systems	McGill	Fall '25, '24	Hong Guo
ENGS 20	Introduction to Scientific Computing	Dartmouth	Winter '23	Hélène Seroussi

GRADUATE COURSEWORK

McGill: Computational Physics, Observational Techniques of Modern Astrophysics, Galaxies & Cosmology, Radiative Processes, Antennas and Propagation [F26], Advanced Radio Astronomy [W27]

Dartmouth: Math for Machine Learning, General Relativity

CONFERENCES & SUMMER SCHOOLS

15-19 Jun 2026	CHORD Face-to-Face	Marseille, FR
	<i>Contributed talks:</i> “Systematics-aware simulation pipeline” and “Rectilinear slicing / elevation motorization,” on behalf of the beam and intensity mapping science working groups	
9-11 Jun 2026	CASCA Annual General Meeting	Montréal, CA
	<i>Contributed talk:</i> “Systematics-aware forecasts for HI intensity mapping with CHORD”	
25-28 May 2026	Canadian Radio Astronomy Summer Institute and Extended School	Montréal, CA
16-20 Mar 2026	Cosmology in the Alps	Les Diablerets, CH
	<i>Contributed talk:</i> “Systematics-aware forecasting for neutral hydrogen intensity mapping with CHORD”	
21-25 Jul 2025	Dunlap Institute Summer School: Introduction to Radio Astronomy	Toronto, CA
2-6 Jun 2025	Michigan Cosmology Summer School	Ann Arbor, US
6-8 May 2025	Rencontre Annuelle du Centre de recherche en astrophysique du Québec	St-Alexis-des-Monts, CA
	<i>Contributed talk:</i> “Systematics-aware cosmological forecasting for CHORD”	
14 May 2024	Senior Honor Thesis Showcase at Dartmouth College	Hanover, US
	<i>Poster:</i> “Understanding the Gravitationally Lensed Blazar PKS 1830-211”	
22 May 2024	Karen E. Wetterhahn Science Symposium at Dartmouth College	Hanover, US
	<i>Poster:</i> “Understanding the Gravitationally Lensed Blazar PKS 1830-211”	
22 May 2024	Christopher Reed Science Competition at Dartmouth College	Hanover, US
	<i>Poster:</i> “Understanding the Gravitationally Lensed Blazar PKS 1830-211”	
7-11 Jan 2024	243rd Meeting of the American Astronomical Society	New Orleans, US
	<i>Poster:</i> “Using a Natural Cosmic Telescope to Study a Distant Radio Quasar”	
10 Aug 2023	MIT Haystack Observatory REU Symposium	Westford, US
	<i>Poster and talk:</i> “Multi-band and temporal analysis of lensed blazar PKS 1830-211”	
24 May 2023	Karen E. Wetterhahn Science Symposium at Dartmouth College	Hanover, US
	<i>Poster:</i> “Primary neutral helium in the heliosphere: distributions and trajectories”	
12-16 Dec 2022	American Geophysical Union Fall Meeting	Chicago, US
	<i>Poster:</i> “Secondary neutral helium throughout the heliosphere”	
31 Oct–04 Nov 2022	20th Annual International Astrophysics Conference	Santa Fe, US
	<i>Poster:</i> “Secondary Helium Neutrals in the Heliosphere”	
27 May 2021	Karen E. Wetterhahn Science Symposium at Dartmouth College	Hanover, US
	<i>Poster:</i> “Density Functional Theory calculations on flexible polymer moieties to train a machine learning algorithm”	

COMMUNITY ENGAGEMENT & OUTREACH ^{*} *Volunteer* [†] *Paid* [‡] *Selected*

27 Jun 2026	Speaker ^{*‡} in the TSI “Astro 101” talk series; title: “Radio slices of the cosmos”
2 May 2026, 4 Oct 2025, 3 May 2025, 28 Sep 2024	Volunteer [*] at McGill family science outreach fairs
23 Feb 2026	Speaker [*] in the Trottier Space Institute (TSI) Lunch Talk Series
19 Feb 2026, 27 Feb 2025	Speaker ^{*‡} at the TSI and McGill Physics Visit Day Graduate Research Showcase
13 Feb 2026, 15 Nov 2024	Mentor and general volunteer [*] at the McGill Physics Hackathon
13 Nov 2025	Panellist ^{*‡} on the McGill Society of Physics Students graduate school panel
7 Oct 2025	Speaker ^{*‡} at Astronomie en fût / Astronomy on Tap Montréal
Aug 2025–Apr 2026	Co-coordinator [*] of graduate student lunches with TSI seminar speakers
Aug 2025–Aug 2027	Graduate student representative ^{*‡} to the TSI Board
20 Jul 2026, 18 Dec 2025, 16 Jul 2025, 22 Apr 2025	Volunteer [*] at the McGill “World of STEM” high school sessions
8 Jun 2025	Volunteer [*] at AstroFest Montréal
12 May 2025	Judge [*] at the Royal Vale School grade 5-6 science fair
24 Apr 2024	Speaker [*] at the Dartmouth Physics and Astronomy Society research meeting
Sep 2023–Jun 2024	Peer mentor [*] in the Dartmouth Physics and Astronomy Society
Sep 2022–Jun 2024	Peer mentor [*] in the Women In Science Project at Dartmouth College
Jan-Mar 2022	Tutor [†] for multivariable calculus at the Dartmouth Academic Skills Center
Aug 2016–Jun 2020	Tutor [*] for Latin and math at Exeter High School (NH)